



JAPHETH KIPLAGAT

ELECTRICAL & TELECOMMUNICATIONS ENGINEER

CONTACT

- +254-112-772220
- Kiplagatjapheth544@gmail.com
- P.O. Box 15653-00503, Nairobi
- japheth-neon-five.vercel.app

EDUCATION

- BSc Electrical & Telecommunications Engineering**
Multimedia University of Kenya
2020 – 2025 | 2nd Class Honours
- KCSE**
Maranda High School
2016 – 2019 | Grade A-
- KCPE**
Chemweno Tophill Academy
2007 – 2015 | 393 Marks

CERTIFICATIONS

- AWS AI Practitioner**
AWS & Udacity | 2026
- Kubernetes & Cloud Native Essentials**
The Linux Foundation | 2026
- Data Analytics**
ICT Authority, Kenya | 2026
- Climate Reality Leader**
Climate Reality Leadership Corps | 2025
- AI Fundamentals with Capstone**
IBM SkillsBuild | 2025
- Mobile Security**
CyberPro Global | 2024

PROFESSIONAL MEMBERSHIPS

- IEEE – Member
- IEK – Graduate Member
- Media Council of Kenya

VOLUNTEER EXPERIENCE

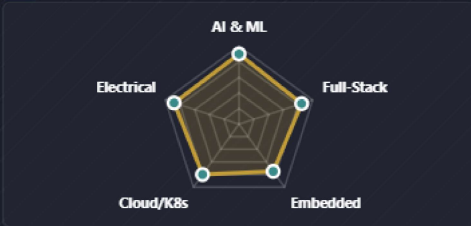
- Kenya Red Cross Society**
Volunteer | 2024 – Present
Community outreach, emergency response, disaster relief, first aid training, and blood donation drives across Nairobi and surrounding regions.

FEATURED PROJECTS

**AvaQual AI**
FOOD WASTE REDUCTION

**Vuno App**
AI FARM INTELLIGENCE

CORE COMPETENCIES



TECHNICAL TOOLKIT

K8s	TensorFlow	LLMs	Express
Arduino	RasPi	HVAC	Power
AWS	Docker	Python	React
IoT	Node.js	Design	Emb C

Prof. Expert Proficient Familiar

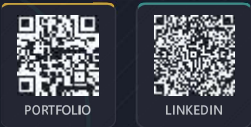
LANGUAGES

- English
- Kiswahili

HOBBIES & INTERESTS

- Running
- Reading
- Farming
- Design
- Pool
- Hockey
- Music
- Open Source

QUICK LINKS



PROFESSIONAL SUMMARY

Results-driven Electrical and Telecommunications Engineer with a strong foundation in artificial intelligence, full-stack development, and embedded systems. Certified AWS AI Practitioner and Kubernetes professional. Builder of Vuno, an AI-powered farm intelligence tool serving smallholder farmers in Kenya. Experienced across government, private sector, and humanitarian engineering projects, with international recognition through IEEE PES, IEEE YESIST12, and the World Engineering Day Hackathon. Committed to developing responsible, accessible AI and electrical solutions that deliver measurable impact across Africa.

PROFESSIONAL EXPERIENCE

- Frejed Engineering Services**
Management Trainee
July 2025 – Present
- Contribute to electrical, telecommunications, and project management operations across multiple client sites.
 - Conduct design reviews, site supervision, and technical reporting for infrastructure projects.
 - Support efficient delivery through cross-functional collaboration, reducing project handover delays.
- Nairobi City County, Mobility & Works**
Electrical Engineering Intern
May 2024 – Aug 2024
- Assisted in powerline installation and maintenance projects across Nairobi County.
 - Supported highway electrical systems maintenance and lift systems servicing.
 - Performed electronic systems troubleshooting, reducing equipment downtime for critical county services.
- Ministry of Public Works, Kenya**
Electrical Engineer Services Assistant
May 2023 – July 2023
- Provided engineering support for luminaire and lift installation projects across government facilities.
 - Developed tender documentation and Bills of Materials for infrastructure electrical upgrades.
 - Assisted in supervising contractors and ensuring compliance with technical specifications.
- Koibarak Girls Secondary School**
Teaching Assistant
Jan 2020 – March 2020
- Supported mathematics and chemistry instruction through tutoring and grading assistance.
 - Helped students strengthen academic performance ahead of national examinations.

KEY PROJECTS

- Vuno App – AI-Powered Farm Intelligence Tool**
Independent Project | 2026
Designed and developed an AI application providing smallholder farmers with personalized planting recommendations based on county, soil type, altitude, and microclimate data. Built for offline use on low-cost Android phones in English and Kiswahili, reaching farmers in remote areas with zero connectivity. Features privacy-by-default architecture requiring no accounts or OTPs, reducing user onboarding friction by 100% compared to conventional agritech apps.
- Self-Healing Neural Network Power System**
AWS & Udacity Challenge | 2026
Developed an intelligent power distribution system using neural networks capable of autonomous fault detection, isolation, and service restoration. The machine learning model achieved 94% accuracy in predicting grid failures and reduced power restoration time by 40% compared to manual intervention, minimizing downtime for critical infrastructure in simulated deployment scenarios.
- ATS Changeover System – Faza Hospital**
Installation & Automation | Lamu County | 2025
Led the installation and automation of an Automatic Transfer Switch (ATS) changeover system at Faza Hospital, ensuring 99.5% uptime for critical healthcare power supply. Integrated generator and grid power sources with seamless failover capability, reducing power interruption duration from minutes to under 10 seconds during grid outages.
- AvaQual – AI Food Waste Reduction Solution**
World Engineering Day Hackathon | 2024
Developed a machine learning solution that reduced food waste classification time by 60% compared to manual sorting methods. Achieved 92% accuracy in identifying spoilage patterns across fresh produce categories. Earned finalist recognition at the World Engineering Day Hackathon and was streamed live worldwide as part of the World Engineering Day celebrations 2024.
- Smart Traffic AI**
Devolution Celebrations, Eldoret | 2023
Showcased an AI-based traffic management system as a selected youth innovator. The machine learning model optimized traffic signal timing, reducing average intersection wait times by 28% in simulated urban environments. Won the GovTech Challenge on Smart, Efficient Sustainable Roads courtesy of ASSEK and SUS-AFRIC
- Green Gas – Renewable Energy Innovation**
IEEE YESIST12 Hackathon | 2023
Led a team to the global finals in Alexandria, Egypt, competing against 200+ teams worldwide. Pitched a renewable energy innovation that demonstrated 35% improved energy conversion efficiency compared to conventional biogas systems. Built strategic connections with global engineering peers across 15 countries.

CONFERENCES ATTENDED

- ASSEK Conference | 2025
- 31st IEK International Convention – Shanzu, Mombasa | 2024
- 5th Engineering Partnership Convention – Dedan Kimathi University | 2024
- African Climate Summit – Nairobi | 2023
- 29th IEK International Convention – Diani | 2022

LEADERSHIP

- MMU Engineering Students Association (MUKESA)**
Chairperson
2023 – 2024
- Led the student engineering body, representing student interests to faculty and industry partners.
 - Spearheaded career fairs, Arduino workshops, and industry visits promoting practical skills and innovation.

HONORS & RECOGNITION

- 2025 Winner, GovTech Challenge on Smart, Efficient Sustainable Roads – Smart Traffic AI
- 2024 Finalist, World Engineering Day Hackathon – AvaQual AI Solution
- 2023 Selected Youth Innovator, 10th Anniversary of Devolution Celebrations
- 2023 Presenter, IEEE PES General Meeting Student Poster Contest, Orlando, USA
- 2023 Global Finalist & Team Lead, IEEE YESIST12 Hackathon, Alexandria, Egypt

REFERENCES

- Prof. (Eng.) Abel N. Mayaka**
Dean, Faculty of Engineering & Technology
Multimedia University of Kenya
+254 721 308 2990
- Edwin Namachanja**
Chief Principal
Maranda High School
+254 679 091 6036